When Search Is Not Enough: Diagnosing Failure Before Constructing Verified Reasoning Capabilities

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CORA on ARC-AGI-2 · development results · ARC Prize Research Summit, 23 October 2026

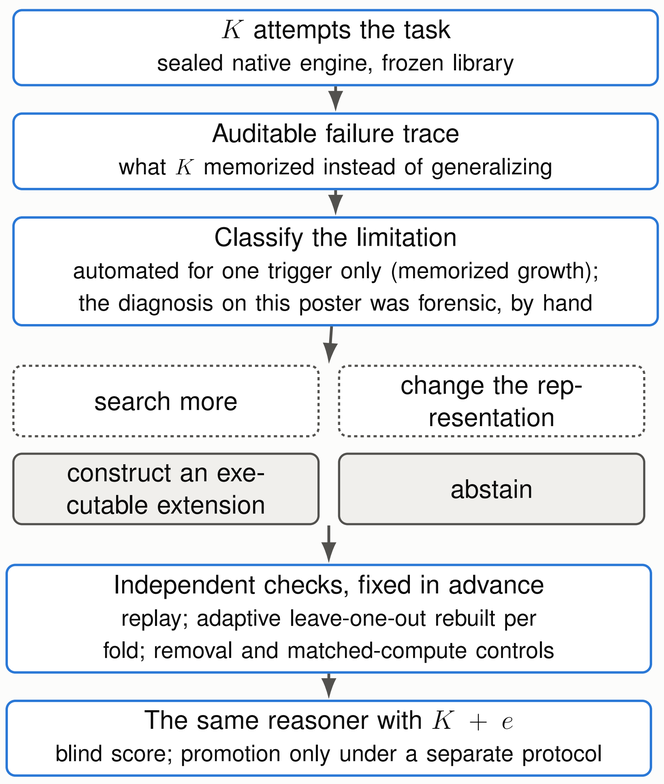
The question

How can a reasoning system know when its existing way of solving problems is inadequate, and acquire a better one without compromising the evidence used to judge it?

A failed solve can need different responses:

- **search correction**: the answer is expressible, search missed it
- **representation correction**: the needed view was screened out
- **executable capability construction**: the language lacks it
- **evidence insufficiency**: the examples underdetermine the rule
- **certification failure**: a valid candidate fails the checks

The CORA loop

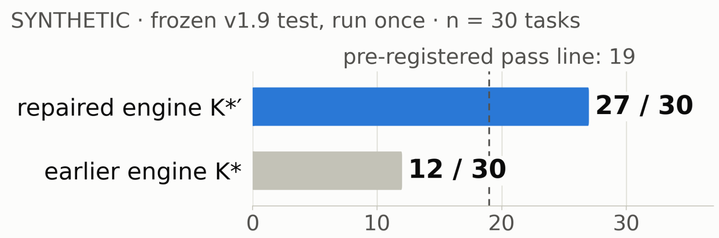


Solid: implemented and measured. Dashed: proposed, not automated. A general failure-type controller is **proposed, not implemented**.

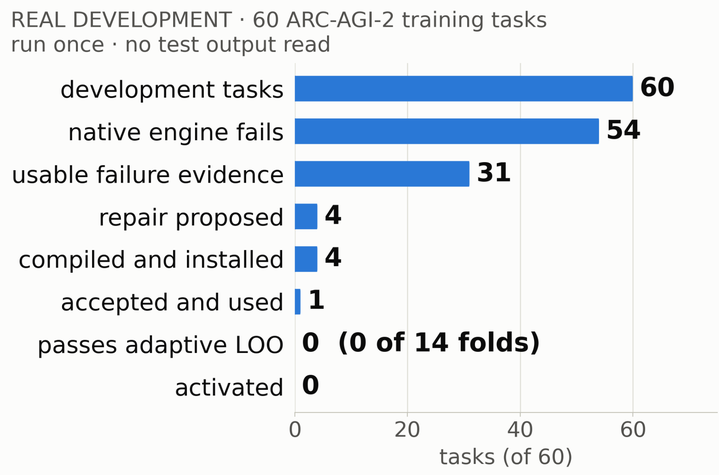
Four claims, four kinds of evidence

(a) the extension causes the solve	development only: nothing accepted without it, or with matched compute; not certified
(b) the real failure trace matters	demonstrations-only and shuffled failures never gave an accepted, used extension (4 tasks)
(c) a bounded semantic extension	**not established**
(d) retention and later reuse	**untested**

What has been measured

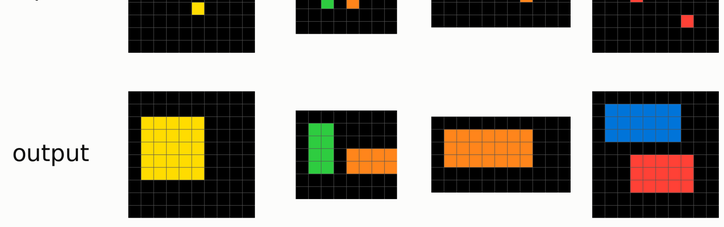


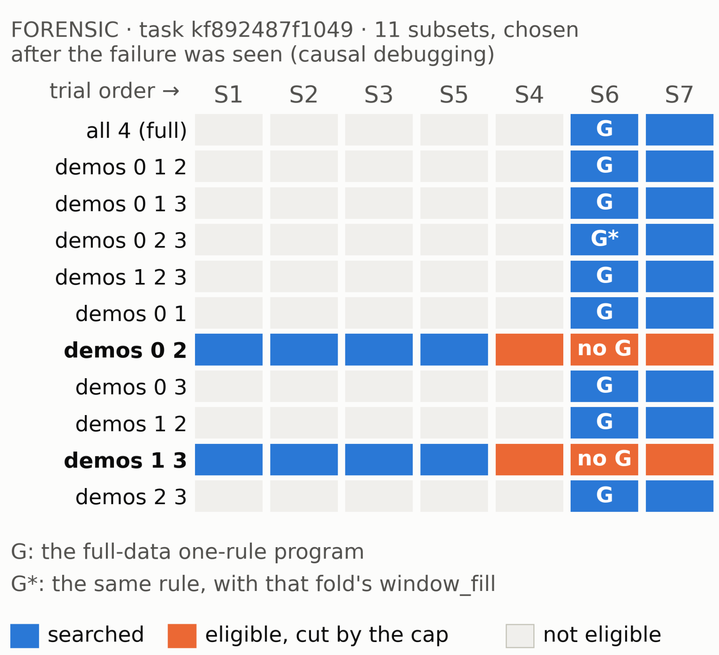
A synthetic claim only. The repaired engine also accepted every deliberately wrong extension it was given (5 of 5, 28 of 28).



One constructed extension was accepted and used by the unchanged engine; none survived adaptive leave-one-out. **No new ARC solve**: the certified training score stays 185/1,000.

The right view was cut, not missing

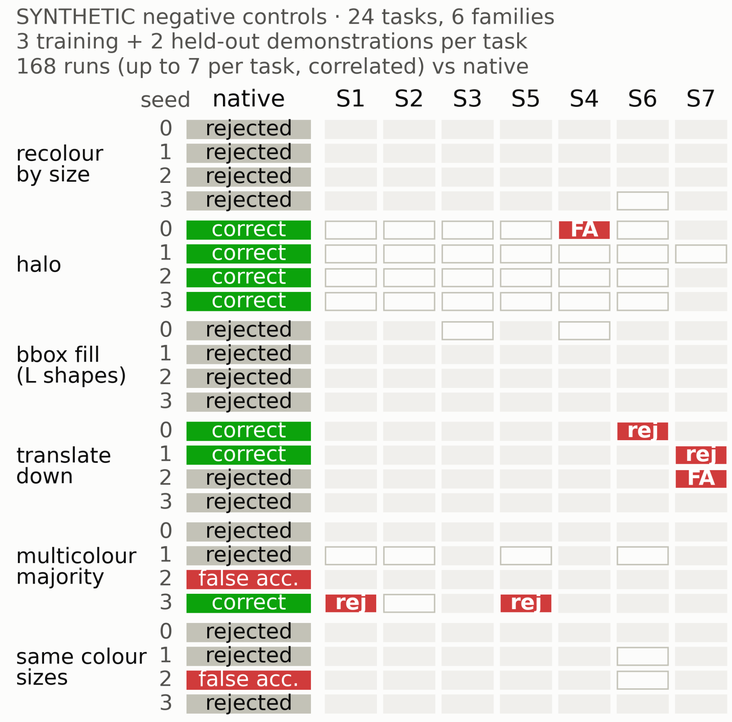




The engine searches at most four eligible views, in a fixed order. With only demonstrations 0 and 2, or 1 and 3, finer views become eligible and fill the four slots. The colour-layer view S6, where G lives, is never searched. Two of the four slots are duplicates: S2 and S5 give the reasoner exactly S1's input there.

Can a neural model pick the view? Not here. In a pre-registered synthetic test (35 cases that needed a different view), the fixed order picked right 35 times; a frozen Qwen2.5-1.5B 19 times, and only once when shown the reasoner's real failure record (a shuffled record did the same); I-JEPA 3 times.

Negative controls before any repair

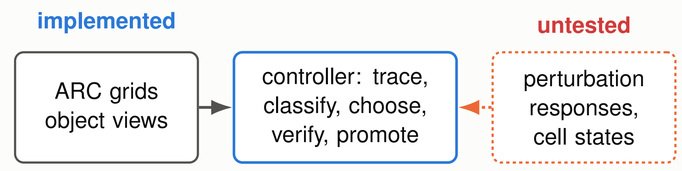


No task gained. Forcing one view to be searched first harmed 5 of 24 tasks: 4 of the 7 natively correct tasks lost their answer, and 1 rejected task became a false acceptance. The variant-first repair we had built was therefore **not run**.

What a valid repair must respect

- Every fold, inner and outer, selects its representation from **its own training demonstrations only**.
- Anchoring a fold to the parent's view would use the fold's held-out demonstration: **forensic only**, never a certificate.
- Removing duplicate views reaches S6 on both failing pairs, from the fold's data alone. Tested on 41 synthetic tasks, it gained 1 and harmed 4 (2 correct answers lost, 2 false acceptances) through the coarse views it let in: **rejected**.
- Observed, not a theorem: removing a demonstration made an eligible view ineligible in 7 of 724 development comparisons (pooled coverage).

A portable controller? BIOLOGICAL APPLICATION: UNTESTED



The hypothesis: the controller and its evidence and certificate interfaces are portable; ARC operators are not. A test needs typed adapters, training-only diagnosis, held-out perturbations and cell states, and confounder controls. No biological data or result exists here.

Not claimed · evidence

No ARC-AGI-2 solve or score gain; no semantic invention; no automatic failure-type controller; no domain generality; no results from the separate VDCG project. The 120-task pilot is frozen and **has not been run**.

Every number, with commit and hash: `records/CORA_SUMMIT_POSTER_EVIDENCE_20261012.md`; figure hashes: `figures/PROVENANCE.json`.